

Intelligence

There are many theories about how intelligence evolved, what it actually constitutes, and which factors are key to high intelligence.

What is intelligence?

Intelligence is our ability to acquire information from our surroundings, incorporate that information into a knowledge base, and then apply it to new situations and contexts. While there are many models for how human intelligence evolved, language and social living undoubtedly played a role as this enabled knowledge to be passed on from generation to generation. The evolution of human intelligence has led to our success as a species, enabling us to adapt to and inhabit almost all environments on Earth.

THERE ARE OVER
**1,000 HUMAN
GENES THAT
HAVE BEEN LINKED
TO INTELLIGENCE**



1 Acquire
Information is gathered through various experiences, understood, and retained for processing.



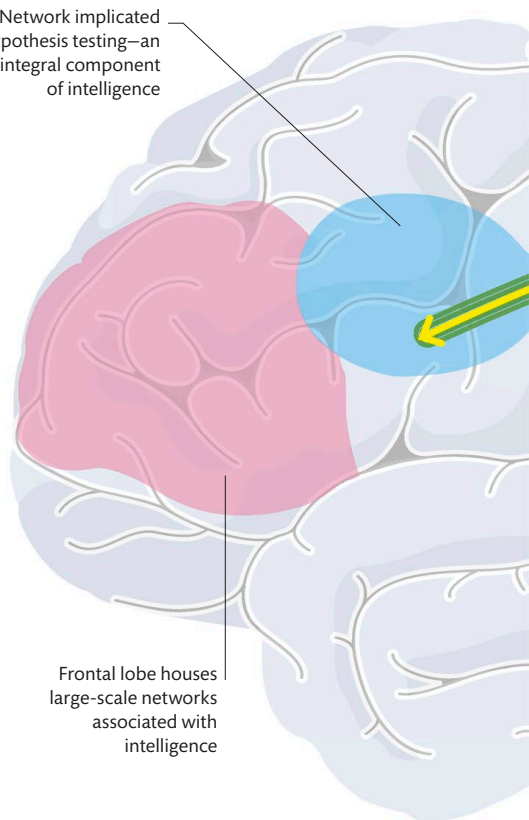
2 Process
New information is critically analyzed, compared with existing knowledge and placed in context.



3 Apply
Existing knowledge is applied to a new situation or problem, as opposed to being repeated from memory.



Network implicated in hypothesis testing—an integral component of intelligence



Frontal lobe houses large-scale networks associated with intelligence

Theories of intelligence

Some studies suggest that connectivity between the prefrontal and parietal cortices and small areas of neurons (networks) is the key to high intelligence (above). Other explanations (right) have also been put forward, suggesting that intelligence is related to connectivity across the brain as a whole.

Types of intelligences

Intelligence is often spoken of in a broad sense, but there is a theory that multiple intelligences exist. It recognizes that people may have the capacity to acquire and apply knowledge in specific areas. For example, someone may struggle with solving math problems but can reproduce a piece of music after hearing it only once. Some argue this theory supports a more realistic definition of intelligence, while critics claim that these “intelligences” are merely aptitudes.



Naturalist

Recognizes features of plants and animals and infers insights based on what is known about the natural world.



Musical

Sensitive to rhythm, pitch, tone, melody, and timbre and applies this to playing and composing music.



Logical-mathematical

Quick with numbers and easily quantifies things. Figures out problems systematically and thinks critically about issues.



Existential

Uses observations, insight, and knowledge to explain the external world and the role of humans in it.



Interpersonal

Sensitive to people's moods, feelings, and motivations. Applies this to relationships and helping groups function.



Bodily-kinesthetic

Uses heightened body awareness, coordination, and timing to master physical activities such as sports.